

Received June 22, 2017, accepted September 29, 2017, date of publication October 27, 2017, date of current version February 14, 2018.

Digital Object Identifier 10.1109/ACCESS.2017.2767059

Analysis of Message Attacks in Aviation Data-Link Communication

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This work was supported in part by the National Key Research and Development Program of China under Grant 2016YFB0800404 and in part the National Natural Science Foundation of China under Grant U1433105, Grant U16361112, and Grant U1636212.

ABSTRACT In aviation communication system, VHF data-link message is used to transmit flight status, airport control instructions as well as the other sensitive information. Radio-transmitted information can be intercepted or tampered with, leading to the disclosure of private data, illegal control, and hijacking. This paper analyzed three attack routes of monitoring, entity camouflage, and man-in-the-middle attack. Man-in-the-middle attack was also able to work in next generation aviation communication networks. Two viable experiments were designed to test data-link message attacks including information leakage and the entity camouflage attack. These experiments may also be used as penetration tests of secure of aviation communication system.

INDEX TERMS Message attack, monitoring, entity camouflage, man-in-the-middle (MITM), aviation.

I. INTRODUCTION

The latest famous attack event of aviation communication system occurred on July 29, 2016. BBC news reported the “Vietnam airport screens hacked event”. “Screens and sound systems at Hanoi and Ho Chi Minh City airports broadcast anti-Vietnamese and Philippines slogans” [1]. In the current civil aviation communication systems, the core of the ground system is usually physically isolated from the Internet, hackers can hardly have any direct contact with the core system.

Some experts believed that, for aviation communication system, as a part of ground-air communication of civil aviation, many message vulnerabilities may lead to even greater losses. In 2009, Steve Jobs’ private jet was tracked over the internet because of ACARS (aircraft communication addressing and reporting system) message leakage [2]. In the Hacker Conference 2013, a security consultant demonstrated the hijacking process through vulnerabilities of ADS-B and ACARS protocols using an Android mobile application [3], but the “FAA and EASA said hijacking planes using an app is not possible” [4].

These two security events belong to different kinds of data-link message attacks: monitoring and entity camouflage. For monitoring attack, attacker may monitor messages using any radio or PC with transceiving equipment to get flight number, departure airport, destination airport and other

information, because current ground-air communication information is unencrypted. For entity camouflage attack, attacker can camouflage as a ground station and send an illegal message to the aircraft, resulting in airborne communication system disorders, even manipulating the aircraft, because there is currently no authentication between flight and ground station.

In this paper, we focused on studying three kinds of attacks: monitoring, entity camouflage and Man In The Middle (MITM) attack. For the former two, we analyze attack routes and construct two experiments that can be used to test aviation communication security. For the third, we only analyze the attack route in next generation aviation communication networks (ATN), because ATN is still not implemented.

II. RELATED WORK

Current aviation communication follows ARINC Specification 620 [16], which defines the application messages and used by xxx number of airports/airlines. ARINC 618 [17] specifies the protocol used between the CMU (Communication Management Unit) and the RGS (Remote Ground Station). The ARINC series uplink and downlink communication procedure is shown in Fig. 1.

For uplink communication process, the client’s terminal send first the message, that will be sent to the aircraft, to

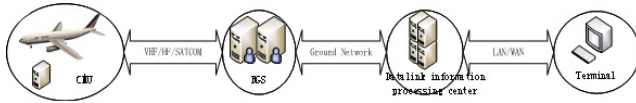


FIGURE 1. Uplink and downlink procedure.

the data-link processing center for analysis through the LAN or the WAN. After analysis, the message will be sent to the remote ground station (RGS) through the ground network, the receiving station will then send the message to the airborne receiver.

For downlink communication process, the aircraft sends message by the airborne transceiver, the RGS then forward the received signals via ground network to the ground data-link information processing center for further analysis, and the message processed in real-time are then forwarded to the clients through a LAN or a WAN, so that relevant entities will have real-time monitoring of the aircraft's running status.

The security flaws of the message system such as information leakage and weak entity authentication. The ARINC 823 protocol [18] pointed out the existence of data leakage, data fraud, entity camouflage and denial of service attacks and other security threats of ACARS system, and also proposed digital certificate, encryption and digital signature technology for AMS (ACARS MESSAGE SECURITY). However, it is still unclear about how these vulnerabilities of ACARS are used for attacks. The simulation platforms are expensive and complex. For example, the USAF (United States Air Force) built a Reconfigurable Cockpit and Avionics Testbed (RCAT) around 2000 [12]. It is obviously unsuitable for attacks experiments.

Many papers discussed advanced security schemes using cryptographic algorithms, but these advanced security schemes hardly work in ACARS system, unless the next generation air transportation system will replace the current system [12], [8]. For example, for vulnerabilities of AMS entity authentication and other aspects, some scholars researched it from a theoretical perspective when designing solutions with enhanced security [5], [7]–[10], but these encryption and digital signature algorithms cannot resist entity camouflage without CA (Certificate Authority), according to the paper [20]. As stated in reference [19], IP ATN (Internet-protocol based aeronautical telecommunication network), the next-generation aviation communications system, uses ECC, AES encryption algorithm and PKI to resist message leakage and entity camouflage. However, it will take a long time to be implemented in civil aviation. Security of ATN system needs a further discussion.

Gradually, both ADS-B (Automatic Dependent Surveillance - Broadcast) and ACARS will be used in air transportation, which are similar in data-link communication process and security problems. Paper [6] pointed out that unauthenticated ADS-B data message may lead to system invasion by disguised planes, and they provided a batch-certification solution based on identity signature. Paper [14] evaluated

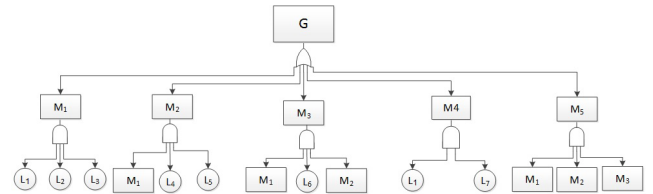


FIGURE 2. Data-link message attack tree. M1: monitoring attack; M2: data forgery attack; M3: entity camouflage attack; M4: illegal invasion; M5: Man-in-the-middle; L1: search frequency point; L2: monitor message; L3: analysis message format; L4: construct fake message; L5: put fake message into communication channel; L6: replay entity identity information; L7: input mess data into communication channel.

FFX encryption algorithm in ADS-B environment to prevent interception. Paper [15] described a classification of attacks and examined the potential damage that the attacks may have on air transportation operations. The classification helped provide a comprehensive understanding of the threats associated with the ADS-B implementation, thereby supporting risk analysis and risk management efforts. Paper [13] summarized several typical attacks on ADS-B: Aircraft Reconnaissance, Ground station flood denial, aircraft flood denial, ground station target ghost injection/flooding, aircraft target ghost injection/flooding, virtual aircraft hijacking, virtual trajectory modification, aircraft disappearance, aircraft spoofing, they proposed broadcast authentication and location verification schemes. In reference [11], ADS-B data-link vulnerability has been analyzed from the perspective of air service, including: Ground Station Flooding, Ghost Aircraft Injection/Flooding, Aircraft Disappearance, Virtual Trajectory Modification/False Alarm Attack, and Aircraft Spoofing.

III. ANALYSIS OF TYPICAL ATTACKS

In this section, we discuss five strategies data-link attacks such as monitoring, data forgery, entity camouflage, illegal invasion and man-in-the-middle (MITM).

A. MESSAGE ATTACK TREE

We generate an message attack tree according to possible attacks that may occur during message transmission, as shown in Fig. 2. M represents an attack and L represents an atomic operation which the attacker generates. The connection between the nodes include 'and'  and 'or' . The target of attack threatening data-link security is abstracted as the root node G. Target G can be reached in five ways: monitoring (data leakage), data forgery (data spoofing), entity camouflage, illegal invasion and MITM.

We simply assume that each kind of attack works individually in this paper. The hybrid attack with all kinds of attack doesn't be discussed.

1) THE PATH TO G THROUGH M₁

Figures that are meant to appear in color, or shades of black/gray. Such figures may include photographs,

illustrations, multicolor graphs, and flowcharts.

$$L_1 \& L_2 \& L_3 \rightarrow M_1 \rightarrow G \quad (1)$$

The attack target (G) is privacy leakage. Attacker turns on the monitoring equipment and adjusts the frequency of monitoring equipment (L_1) to capture messages sent by the transmission terminal (L_2), analyses message format (L_3). These messages are transmitted at a fixed frequency in the form of plaintext. Thus, an attacker can monitor the messages using any simple transceiver device. Passenger information including transit information and identity can be stolen.

2) THE PATH TO G THROUGH M_2

$$M_1 \& L_4 \& L_5 \rightarrow M_2 \rightarrow G \quad (2)$$

Attacker has to learn data format by monitoring (M_1), then construct forged data (L_4), and put forged data into communication channel (L_5). There may be two attack targets (G): by sending forged uplink messages to a mislead pilot, or by sending forged downlink messages to hide the flight status.

3) THE PATH TO G THROUGH M_3

$$M_1 \& L_5 \& M_2 \rightarrow M_3 \rightarrow G \quad (3)$$

Attacker forges data by monitoring data (M_1), puts fake data into communication channel (L_5), and communicates as a legal object (M_2). The attack target (G) is to disguise as a ground station to communicate with a flight, or to disguise as a flight to communicate with ground station. In the receiving process of ARCARS messages, there aren't entity authentications between communication parties. The airborne MU only checks CRC and flight registration number in uplink messages. Thus, entity camouflage attack may be implement. Pre-Departure Clearance (PDC) implements clearance before departure clearance permission communication between the pilot and the tower controller through data-link, to replace or partly replace the current Voice Clearance. An attacker could conduct entity camouflage attack in PDC implementation, causing a mess of airport.

4) THE PATH TO G THROUGH M_4

$$L_1 \& L_7 \rightarrow M_4 \rightarrow G \quad (4)$$

Attackers searches communication frequency point (L_1) to input false data (L_7). The attack target (G) is to cause mess of air-ground communications. This attack is easy to succeed by interference [21].

5) THE PATH TO G THROUGH M_5

$$M_1 \& M_2 \& M_3 \rightarrow M_5 \rightarrow G \quad (5)$$

Attacker has to learn data format by monitoring (M_1), then construct forged data (L_4), and put forged data into communication channel (L_5). There may be two attack targets (G): by sending forged uplink messages to a mislead pilot, or by sending forged downlink messages to hide the flight status.

B. MONITORING ATTACK FOR TRANSFER PASSENGERS INFORMATION

With data-link communication system in place, the crew can acquire transfer passengers information and status of the connecting flights on board, used to inform the transfer passengers of different destinations beforehand. The actual process of transfer passenger's information service is: before the flight is landed, ground station sends transfer information and upload it to the aircraft through the data-link. The aircraft automatically receives and informs the cabin attendants about the number of transfer passengers and their respective baggage carousel number, seat number, destination and gate number, so as to notify them in advance. Alternatively, the information management system in the cabin automatically analyzes the packet received from ground station, and displays the information directly on the entertainment display in front of passengers [23].

Because of data-link messages are transmitted in plain text, data leakage is possible. Monitor attack is shown in Fig. 3. The attack steps are as follows.

Step 1: Automatically searches for a frequency point of the communication between the ground station and the aircraft.

Step 2: Keeps monitoring the data-link messages and modulating/demodulating.

Step 3: Looks for transfer passengers' information in the messages using decoder & encoder module with messages specification database.

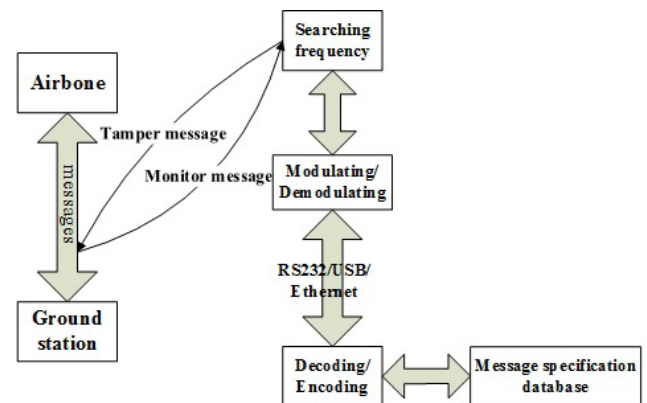


FIGURE 3. Process of monitoring attack.

By monitoring, the attacker is able to gain much information to make other attack. For example, attacker monitors data messages and gets entity identity information to practice entity camouflage attack.

C. ENTITY CAMOUFLAGE ATTACK FOR PDC

At any busy international airport, the air traffic controller sends a clearance permission to the pilot on an aircraft simply with a few keystrokes, lessening the time of waiting that the traditional audio frequency entails.

According to ARINC623 [22], 20 minutes before the aircraft's departure, a PDC application is sent through the DCL/PDC service. ARINC623 protocol has defined

four messages for pre-departure clearance (PDC) to complete the entire process, which are: Departure Clearance Downlink (RCD), Departure Clearance Uplink (CLD), Clearance Readback Downlink (CDA) and the Flight System Uplink message (FSM). We describe the takeoff clearance message flow as Fig. 4.

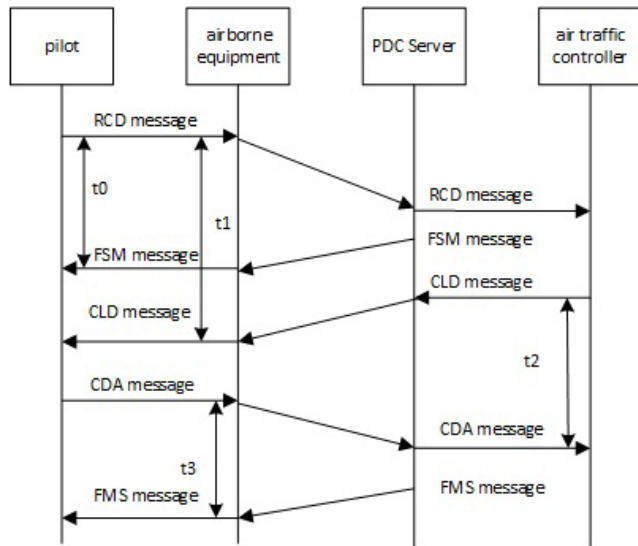
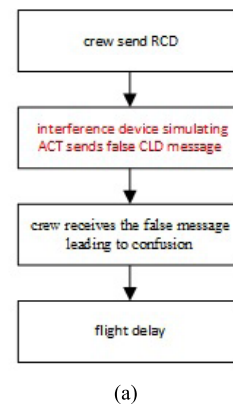


FIGURE 4. The takeoff clearance message flow. [1] RCD - Departure clearance downlink sent by the pilot. That contains flight number, tail number, Automatic Terminal Information Service (ATIS) code, plane model, airport code of departure and airport code of destination. [2] CLD - Departure Clearance Uplink sent by the air traffic controller after approval. [3] CDA - Clearance Readback Downlink sent by the pilot for confirmation. [4] FSM - Flight System Message, the RCD / CDA confirmation uplink packet sent by the digital pre-departure clearance service automatically. [5] t0 - The maximum time interval between the pilot sending RCD and receiving FSM packet. [6] t1 - The maximum time interval between the pilot sending RCD and receiving CLD packet. [7] t2 - The maximum time interval between the air traffic controller sending CLD and receiving CDA packet. [8] t3 - The maximum time interval between the pilot sending CDA and receiving FSM packet.

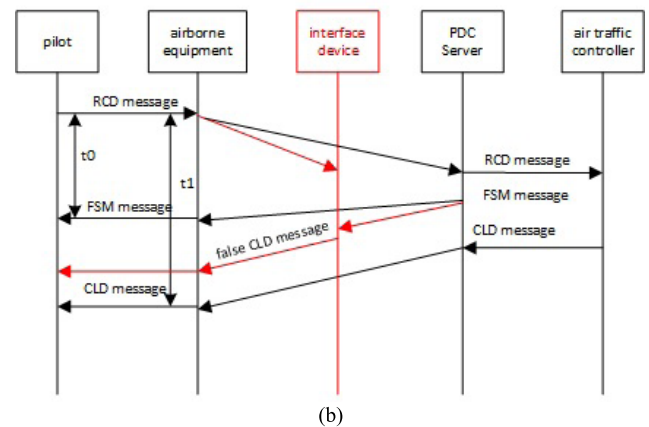
1) ATTACK ROUTE1

If PDC is interfered with, the normal order of clearance will be disrupted. Due to the explosive growth in flights, limited very-high-frequency (VHF) channels are becoming crowded in the control terminal area. Any interference to PDC will directly result in substantial drop in departure clearance efficiency, increasing the workload of controllers and pilots, which could also lead to extensive delays on flights, affecting the normal order of navigation.

According to the takeoff clearance data flow as Fig. 5, if an interfering device sends a simulated RCD message before the aircraft does so, the controller will mistake the aircraft for being ready to take off, thus the flight order is disrupted. If a simulated CLD message is sent after the crew has sent the RCD, the normal clearance service process is disrupted and clearance efficiency reduced. If an interfering device responds an RCD request sent by a crew before the controller could do so, the crew would mistakenly think their clearance request was confirmed, which would then disrupt



(a)



(b)

FIGURE 5. Route1 of entity camouflage attack.

the clearance. When a CLD process is completed, a false CDA would make the controller mistakenly think that the aircraft was ready to take off.

We can use the path3 in Section3.1 to attack PDC service as Fig. 5. When the aircraft has sent a regular departure clearance downlink (RCD), the attacker first listen to the message through the monitor device, before the air traffic control system (ATC) returns the normal departure clearance downlink (CLD). The attacker can send a false CLD message to the aircraft, making it impossible to complete the normal process, delaying the time of departure.

The attack steps are as follows.

Step 1: Conduct monitoring using interference devices.

Step 2: The RCD is being monitored by the interference device when the crew sends an RCD to the PDC server.

Step 3: The interference device sends a simulated CLD packet, containing false information.

Step 4: The crew receives the simulated CLD packet, which leads to the misjudgment of the pilot.

2) ATTACK ROUTE2

On the other way, we can use the path3 to attack PDC service as Fig. 6. Before the aircraft sends an RCD, attacker sends a

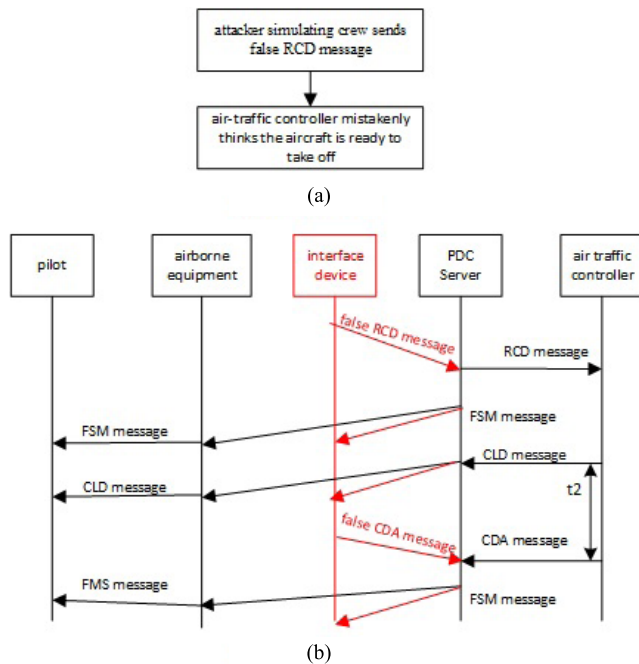


FIGURE 6. Route2 of entity camouflage attack.

simulated RCD to the ATC system, to start the pre-departure clearance process, which leads to the misjudgment of the air-traffic controller that the aircraft is ready to take off, hence disrupting the air-traffic control.

The attack steps are as follows.

Step 1: Conduct monitoring using interference devices.

Step 2: The interference device sends a simulated RCD to the PDC server before the aircraft sends an RCD.

Step 3: The air-traffic controller mistook it for a RCD from an aircraft, and sends a CLD packet accordingly.

Step 4: The interference device receives the CLD packet, and responds with a simulated CDA packet.

Step 5: The air-traffic controller mistakenly read/informed that the aircraft is ready to take off.

D. MITM (MAN-IN-THE-MIDDLE) ATTACK FOR ATN

According to paper [27], [26], in next generation ATN, some treasures such as PKI, digital certification and ECDH key agreement protocol are used to protect monitoring and entity from camouflage attack. The process is shown in Fig. 7. By analysis, we learn that the secretary of parameter Z is the key to keep security of the communication between Airborne A and ground station G.

According paper [28], [29], there are some leaks of digital certification verification in SSL protocol. That is, attacker can forge a self-signed certificate and spoof receiver trusting it. ATN specifications use online certification and similar certification verification process. MITM attack may succeed as Fig. 8.

Suppose the communications between airborne A, ground station G and CA authority were monitored by attacker H. H got *CertA* (digital certification of A), *CertG* (digital certification of G), *request(AirborneID, ...)*,

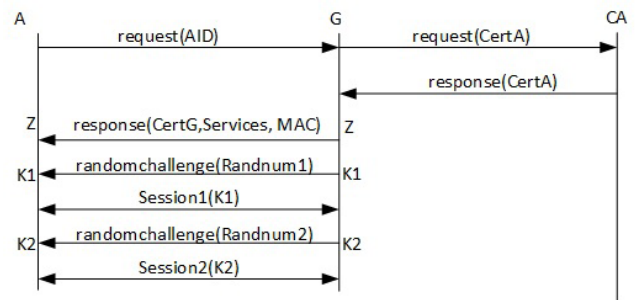


FIGURE 7. ATN secure communication between airborne and ground station.

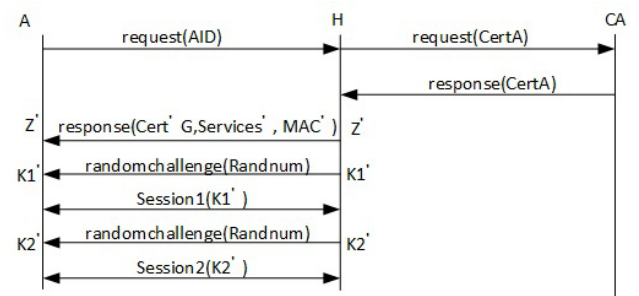


FIGURE 8. Route of Man-in-the-Middle attack.

TABLE 1. Devices for monitoring experiment.

Device type	Device name	Remark
Hardware	Radio or aviation handset	With aviation frequencies
	Data cable	With one end connected to the MIC on the PC, and the other end connected to the aviation handset speaker
Software	PC ACARSD, PLANEPLOTTER and KG-ACARS	With monitoring software For instance, ACARSD is a free software simple to use. The downloading site is www.acarsd.org/
Environment	Any messages in transmission can be monitored	

response(*CertG*, *Services*, *MAC_{KCM}*). Then H computes *Z'* using his own private key *privatekey H* and A's public key *publickeyA*, and sends self-signed certificate, fake services and MAC to A. If A is cheated to trust the self-signed certificate, A will use private key *privatekey A* and H's public key *publickeyH* to compute *Z'*. Thereby, MITM attack succeeds in air-ground communication.

IV. EXPERIMENTAL VERIFICATION

A. MONITORING

The data-link message is transmitted in plain text on public frequency. Hence, simulation attack through message monitoring can be conducted taking advantage of this vulnerability of data-link. Software and hardware devices involved include as Table I.

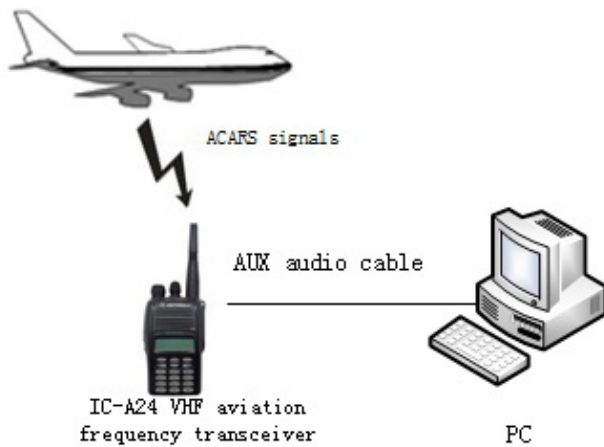


FIGURE 9. Monitoring experiment disposition.

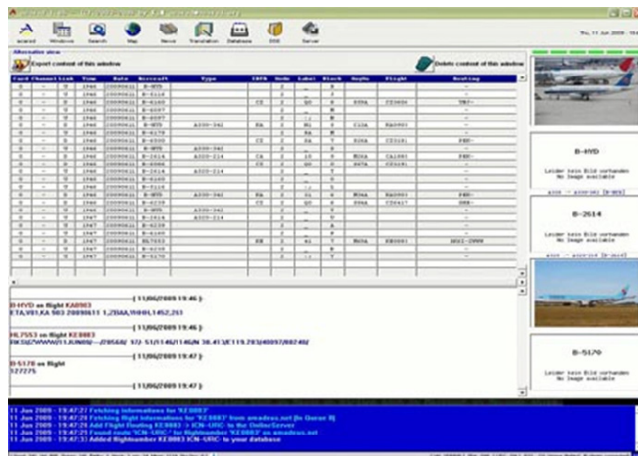


FIGURE 10. Result of monitoring experiment. VHF range is between 118.000 and 135.975MHZ in accordance with the uniform provision of the International Civil Aviation Organization, every 25KHZ is a channel, and 720 channels can be set up for the use of air-ground communication.

We make monitoring experiment of ACARS data-link messages, using a transceiver at aviation frequency. The ACARS signals are received and transmit into a PC. The experiment disposition is as Fig. 9.

For VHF aviation frequency transceiver we use the IC-A24 model of Icom Corp. The transceiver is connected to the PC with AUX audio cable, to send and receive inter-actively. Relevant simulation software such as ASARSD and corresponding decoder shall be installed in the PC. The simulation system is to complete the following tasks: Simulation of the ground-air communication platform, and realization of monitoring analysis of the ground-air data-link. Real-time message processing of received ACARS signals, realization of relevant decoding, verification, and ARINC618 [25] deciphering functions. The result of monitoring experiment is as Fig. 10.

ACARS signals include OOOI events, information reports such as entering and leaving the port, returning to the port and delays, meteorological reports, engine performance

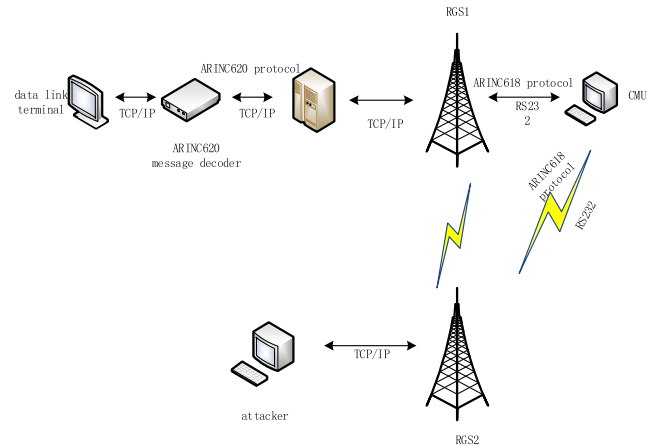


FIGURE 11. Entity camouflage attack experiment disposition. CMU-airborne communication management unit. RGS1-real remote ground station. RGS2- fake remote ground station.

monitoring, fault reports, fuel status reports, selective calls, passenger services, maintenance reports, and other information such as load and balance.

By analyzing all of these monitoring messages, we get transfer passengers private information as follows.

OY-KBI on flight SK0995

0EV033132V

.....

[03/09/2013 11:35]-

B-6076 on flight

/196186933/S/*S

GUO/SHUANGMS 144 11C

ZHAO/CHUNRONG 046 11D

CA/382318720/G/*G

WANG/YULINGMS 067 11H CA/489261441/C

CHU/ZIWEI 049 11J CA/384351170/S/*S

DING/CONG 050 11L CA/112662130/V/*GZHOU/SHIMENG

[03/09/2013 11:35]-

B-6076 on flight

150 12A CA/616606023/C

YANG/HONGPENG 122 12C

CA/411096066/G/*G

GAO/YUANMR 013 12D CA/430693675/C

LIU/JINGMS 012 12H

ZHAO/YIMIN 202 12J CA/604778484/G/*G

WU/JIANQIAN+ 035 12L

CA/133068961/G/*GGE/ZUN

We find that within 5 minutes anyone with a program install and a ebay purchased device in the vicinity of the airport can intercept 15 detailed messages containing passenger transition information, which means over a thousand passengers information could be intercepted within 10 hours.

B. ENTITY CAMOUFLAGE ATTACK

Aviation data-link communication currently does not do entity authentication, the airborne MU only checks CRC and flight registration number on the uplink messages. Therefore, entity camouflage attacks may be conducted based on the

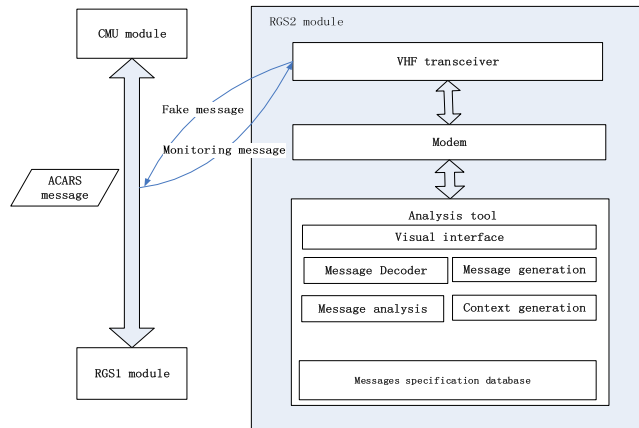


FIGURE 12. Entity camouflage attack simulation. CMU module: Simulate CMU to send the message according to the ARINC 618 protocol to RGS1. RGS1 module: Simulate a ground station to broadcast the message. RGS2 module: Tuning to monitor the uplink message sent by the RGS1 to the CMU, to decode the flight registration information. Replay the flight registration information and send fake messages to CMU module.

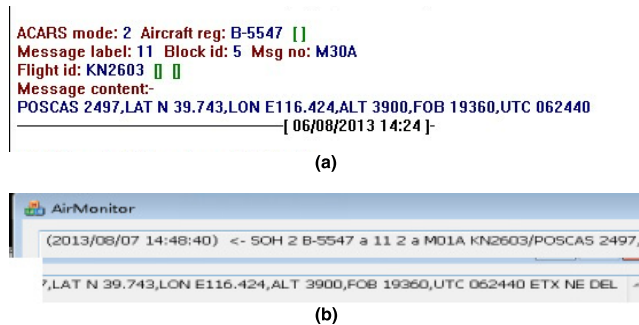


FIGURE 13. Result of entity camouflage attack experiment. (a) Real message including flight registration information that has been intercepted. (b) Fake message.

existing security vulnerability of data-link communication. The experiment disposition is as Fig. 10.

We using OPNET to simulate entity camouflage attack, the process modules are as Fig. 12, and the experiment result are show in Fig. 13.

V. CONCLUSION

Lack of authentication and encryption leads to many security risks in the aviation data-link communication system, as we have found messages are vulnerable to four typical attacks, such as monitoring, data forgery, entity camouflage, and illegal invasion.

In this paper, we analysed the monitoring- and the entity camouflage-attack routes, and constructed attack-simulating experiments. These experiments proved that the two attacks may do harm. The next generation aviation communication network prevents these two attacks by encryption and authentication, but still can't prevent MITM attack. We do not design MITM test because ATN is not implemented.

These experiments can also be used as penetration test cases for air-ground communication system vulnerability

assess. We will develop a kind of penetration test tool with the message attack tree (Setction 3.1).

APPENDIX MONITORED TRANSFER PASSENGERS PRIVATE INFORMATION

OY-KBI on flight SK0995
0EV033132V

[03/09/2013 11:31]-

B-6041 on flight
CHESUSE QNH BELOW TRANSITION
ALTITUDE
WIND 060 DEG 04 MPS 10V100 DEG
VIS 4000
IN HZ
NSC
T 25 /DP 16
QNH 1016 HPA
TRANSITION ALTITUDE 3000 /LEVEL 3600 MU

[03/09/2013 11:33]-

JA832J on flight
/BJSATYA.TI2/ZBAA ARR ATIS B
0331Z EXPECT 36L ILS APPROACH 36R ILS
APPROCH 01 ILS APPROCH
ARR RWY 36L AND 36R AND 01
CTN
REPORT TYPE OF AIRCRAFT WHEN INITIALLY
CONTACT APPROACH
SET XPNDER MODE S ON WHILE TAXR

[03/09/2013 11:33]-

JA832J on flight
WY DEPENDENT PARALLEL APPROACHES
RNAV ARR IS AVAIL

CAVOK
WIND VRB DEG 01 MPS
T 27 /DP 15
QNH 1017 HPA
RPT RECEIPT OF ATIS B ON ZBAAARR5953

[03/09/2013 11:33]-

B-6076 on flight
CREW:5/11/0 CAB:0
WEIGHT DISTRIBUTION
MAX TRAFFIC PAYLOAD 37385
DOW 124732 DOI: 53.69
PAYLOAD 31624 BLKD 2/9
ZFW 156356 MACZFW: 30.45
MZFW 170000
TOF 70883T

[03/09/2013 11:33]-

B-6076 on flight
OW 227239 MACTOW: 2

[03/09/2013 11:33]-

B-6076 on flight
ONG SHENG PEK

[03/09/2013 11:34]-

B-2064 on flight
128475
-----[03/09/2013 11:34]-

B-5106 on flight
0A/4 0B/41 0C/37
LOAD IN CO 359 1/0 2/0 3/359 4/0
0/0
UNDERLOAD BEFORE LMC 9522
LMC TOTAL + -
LDM
ZH9960/03SEP13.B5106.F8Y159.03/06
-SZX.81/1/0.0.PAX/4/78.PAD
-----[03/09/2013 11:35]-

B-6076 on flight
.BJSAGCA 030333
AGM
AN B-6076/MA 142A
- QUBJSAGCA~1RA113031133
AIR CHINA INTERNATIONAL 03-SEP-13
11:32 PAGE1
PASSENGER LIST FOR CA997/03SEP13
ACFT REGN B6076
PEK/YVR
CLASS CZHANG/WEI 038 11A CA
-----[03/09/2013 11:35]-

B-6100 on flight
128475
-----[03/09/2013 11:35]-

B-5106 on flight
128475
-----[03/09/2013 11:35]-

B-6076 on flight
/196186933/S/*S
GUO/SHUANGMS 144 11C
ZHAO/CHUNRONG 046 11D
CA/382318720/G/*G
WANG/YULINGMS 067 11H CA/489261441/C
CHU/ZIWEI 049 11J CA/384351170/S/*S
DING/CONG 050 11L CA/112662130/V/*GZHOU/SHIMENG
-----[03/09/2013 11:35]-

B-6076 on flight
150 12A CA/616606023/C
YANG/HONGPENG 122 12C
CA/411096066/G/*G
GAO/YUANMR 013 12D CA/430693675/C
LIU/JINGMS 012 12H
ZHAO/YIMIN 202 12J CA/604778484/G/*G
WU/JIANQIAN+ 035 12L
CA/133068961/G/*GGE/ZUN
-----[03/09/2013 11:35]-

B-5211 on flight
/BJSATYA.TI2/ZBAA DEP ATIS S
0331Z DEP RWY 36L AND 36R AND 01
CTN
IF FULL RW LGTH NEEDED,ADZ WHEN RQST
ATC CLNC

SET XPNDER MODE S ON WHILE TAX
PDC
RWY INDEPENDENT PARALLEL DEP
36L INITIAL ALT 120036R INITIAL
-----[03/09/2013 11:35]-

B-6076 on flight
ZENG 043 13A CA/553059533/G/*G
YU/SHUYANG 186 13C CA/427349204/C
LI/CHANGSHE+ 054 13D CA/731086285/C
PAN/LIYING 058 13H CA/698235856/C
WANG/RUNXIAO 072 13J CA/121095262/CBU/ZHAOWEN 177
13L CA/6753
-----[03/09/2013 11:35]-

B-5211 on flight
128475
-----[03/09/2013 11:36]-

B-6593 on flight CA1590
INI
-----[03/09/2013 11:36]-

B-5183 on flight
ADCC AUTO WXR UPLINK 2.0
[SA ZBUH] 030300Z SACI63 ZBUH 030300
ZBUH 030300Z 17004MPS 140V240 9000 NSC 23/09 Q1018
NOSIG=
[FC ZBUH] 030118Z FCCI63 ZBUH 030118
ZBUH 030118Z 030312 18005G10MPS 9999 FEW040=
-----[03/09/2013 11:37]-

B-5229 on flight
128475
-----[03/09/2013 11:37]-

B-2869 on flight
FS1/FSM 0337 130903 ZBAA
CXA8116 CDA RECEIVED
CLEARANCE CONFIRMEDB8DC
-----[03/09/2013 11:37]-

B-2869 on flight
128475
-----[03/09/2013 11:37]-

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